SeaBotics software tasks

Work catalogue for the autonomous hull

Goal for the semester: This semester, we will establish a safe and robust software foundation that prepares us for further autonomous development next semester. The boat must be able to enter the correct operating mode, stop safely, be remotely controlled, understand its own position and orientation, and complete a basic autonomous journey between two points. The operator must be able to monitor the boat's status from shore.

Student association:	SeaBotics
Document type:	Work catalogue for the 26-Fall semester
Platform:	Ubuntu Linux 24.04 LTS, NVIDIA Jetson AGX, ROS 2, Luxonis cameras, GNSS–INS and IMU
Last updated:	August 28, 2026

Contents

1 Purpose and scope

This document gathers the software work that should be completed this semester to prepare SeaBotics for further autonomous development next semester. The tasks describe the need, purpose and a clear end result, while allowing the person responsible to investigate and select a suitable solution.

Task allocation is determined by the software lead, Dag Isak. Each task should have a clearly designated owner. The person responsible must ensure the necessary clarifications, integration, testing, documentation and a demonstrable deliverable. Several people may contribute when the scope or dependencies make this appropriate.

Anyone needing technical support, clarification or help with a task is encouraged to contact the software lead. It is better to ask for support early than to let uncertainties or challenges delay the work. Contact information: IsakS on Discord; in an emergency, call +47 991 25 476.

2 Milestones for the semester

Complete obstacle avoidance is not required until safe stopping, positioning, manual control and basic A–B travel have been demonstrated.

Milestone 1: Safe and remotely controllable

Tasks 1–6, 15, 18, 19 and 21.
Demonstration: Change mode safely, stop the thrusters, remotely control the boat and show reliable position and orientation data.

Milestone 2: Autonomous A–B travel

Tasks 7, 8, 16, 17, 20 and 22.
Demonstration: Select a destination, complete an autonomous journey without obstacles, monitor the mission and replay the test data afterwards.

Milestone 3: Understand obstacles

Tasks 9–14.
Demonstration: Detect an obstacle, estimate its position, plan a safe alternative route and follow it in simulation before testing on the water.

3 Dependency overview

Table 1: High-level dependencies between the work areas.

Function	Important prerequisites
Safe manual operation	Safe stopping, operating modes, remote control, communication and heading data
Autonomous A–B travel	Manual operation, position, orientation, destination handling, system health and test recordings
Environmental awareness	Camera images, field data, object detection and obstacle positioning
Obstacle avoidance	A–B travel, environmental awareness, route planning and route following
Reliable development	Foxglove, test recordings, automated checks, GitHub workflow and reproducible environments

4 Safety and basic control

1. Safe stopping of the boat

Need and purpose

The operator must always be able to make the boat safe if communication is lost, the software fails or the boat behaves unexpectedly.

Responsibility for the task

The person responsible must define how the boat enters a safe state, which events trigger a stop, how the stop reaches the thrusters, and how the system confirms that propulsion has actually been disabled.

The task is complete when

- The emergency stop stops all thrusters.
- Loss of required communication produces a safe and predictable response.
- The boat cannot restart without an explicit action by the operator.
- The function has been tested with the mechanical and electronics group.

2. The boat's operating modes

Need and purpose

The boat must know who or what has control so that manual and autonomous control never operate simultaneously.

Responsibility for the task

The person responsible must establish the safe-stop, manual and autonomous modes, including rules for changing modes and the conditions that must be met before a mode can be activated.

The task is complete when

- The boat is always in exactly one mode.
- Invalid or unsafe mode changes are rejected.
- The confirmed mode is available to the rest of the system.
- Critical faults return the boat to a safe state.

3. Visible indication of operating mode

Need and purpose

People nearby must be able to see immediately whether the boat is disabled, remotely controlled or autonomous.

Responsibility for the task

The person responsible must ensure that the signal tower shows the boat's confirmed mode: red for safe stop, yellow for manual and green for autonomous.

The task is complete when

- The signal tower follows the confirmed mode.
- An unknown status or fault cannot be displayed as a safe state.
- All mode changes can be demonstrated reliably.

4. Remote control of the boat

Need and purpose

The boat must be capable of manual control during launching, testing, retrieval and situations in which autonomous operation is not desired.

Responsibility for the task

The person responsible must investigate how a remote controller connects to the boat and how operator input is converted into clear movement commands.

The task is complete when

- The operator can control propulsion, direction and stopping.
- The relationship between input and response is clear.
- Loss of the controller or connection is handled safely.
- Commands are accepted only in manual mode.

5. Stable and easy manual operation

Need and purpose

The operator should be able to ask the boat to travel straight or turn without continuously correcting for wind, current or unequal thruster performance.

Responsibility for the task

The person responsible must investigate how available sensors can help the boat maintain the desired heading and movement.

The task is complete when

- The boat can maintain a selected heading for an agreed period.
- The operator can override or disable the assistance.
- The effect can be compared with operation without assistance.
- The behaviour is documented through tests on land and water.

5 Position, destination and autonomous travel

6. Combined position and orientation

Need and purpose

Navigation needs one reliable description of where the boat is, which way it is pointing and how it is moving.

Responsibility for the task

The person responsible must make information from GNSS–INS, IMU and magnetometer available to the rest of the system and clarify coordinate systems, units, timestamps, update frequency and invalid measurements. The task may be divided if necessary.

The task is complete when

- Position, orientation and relevant movement are published through ROS 2.
- Units and coordinate systems are documented.
- Other components can use the data without sensor-specific details.

- Disconnected sensors and invalid measurements are detected.

7. Destination selection and handling

Need and purpose

Autonomous travel requires a clear mission, not merely knowledge of the boat's position.

Responsibility for the task

The person responsible must define how a destination is specified, represented, changed and cancelled, and how the system determines that the destination has been reached. The first version should support decimal degrees for coordinates and be extensible with visual destinations later.

The task is complete when

- The operator can specify, change and cancel a destination.
- The destination is available to the autonomy software.
- The arrival tolerance is defined.
- Invalid destinations are rejected.

8. Autonomous travel between two points

Need and purpose

The boat must turn its own position and a selected destination into safe and stable movement.

Responsibility for the task

The person responsible must get the boat from a starting position to a specified destination in an obstacle-free area and determine how heading and propulsion are adjusted along the way.

The task is complete when

- The boat can begin autonomous travel towards a valid destination.
- The heading is corrected for drift.
- The boat slows down and stops at the destination.
- Autonomous movement can be cancelled immediately.
- The result is measured using the trajectory and final distance to the destination.

6 Camera, environmental awareness and obstacles

9. Reliable camera images on board

Need and purpose

The software must receive stable camera data before the images can be used for navigation or detection.

Responsibility for the task

The person responsible must achieve stable operation of the Luxonis cameras on the Jetson AGX and make relevant image streams available to the system.

The task is complete when

- The cameras start reliably.
- RGB, stereo and relevant depth data are published through ROS 2.
- The images have correct timestamps and camera frames.
- Faults are reported.
- The load on the onboard computer has been mapped.

10. Dataset from the boat's real environment

Need and purpose

Recognition of buoys, quays and vessels requires images that represent the conditions the boat encounters.

Responsibility for the task

The person responsible must plan and carry out the collection and organisation of field data with variations in light, weather, distance and viewing angle. The material must be suitable for labelling and training.

The task is complete when

- A collection plan exists.
- The dataset covers the agreed categories and conditions.
- The recordings are organised and traceable.
- Storage, privacy and sharing have been clarified.
- Quality has been checked and documented.

11. Detection of relevant objects

Need and purpose

The boat must turn images into information about buoys, quays, vessels and other relevant objects.

Responsibility for the task

The person responsible must investigate and create an initial solution for the agreed object types. The first goal is reliable observation; complete obstacle avoidance comes later.

The task is complete when

- Objects are detected in recordings and live streams.
- The results are published for the rest of the system.
- Each observation includes type, confidence and image location.
- The solution has been evaluated on a separate test set.
- Limitations are documented.

12. Positioning obstacles around the boat

Need and purpose

Finding an object in an image is not enough; the boat must understand whether it is actually in the way.

Responsibility for the task

The person responsible must combine camera observations and distance data into a usable description of obstacles around the boat.

The task is complete when

- Obstacles are assigned an estimated direction and distance.
- The results are expressed in a coordinate system understood by navigation.
- Uncertain measurements are clearly marked.
- The solution has been tested against objects with known positions.

13. Safe route around obstacles

Need and purpose

When an obstacle lies between the boat and the destination, the boat must be able to choose an alternative path.

Responsibility for the task

The person responsible must investigate how a feasible route can be created and updated from the destination, position and known obstacles. The task concerns where the boat should travel, not detailed thruster control.

The task is complete when

- The system creates a route in an agreed test area.
- The route maintains the defined safety distance.
- The route is updated when the situation changes.
- The system can report that no safe route exists.
- The result is demonstrated in simulation before a water test.

14. Following a planned route

Need and purpose

A planned route is valuable only if the boat can follow it reliably.

Responsibility for the task

The person responsible must turn waypoints into continuous movement and handle transitions, deviations, route changes and arrival.

The task is complete when

- The boat follows a route with multiple points.
- Deviations are corrected in a controlled manner.
- The route can be changed while moving.
- The boat stops or requests assistance if the route cannot be followed safely.

7 Communication and operator overview

15. Stable connection between boat and shore

Need and purpose

The operator needs the boat's status and commands without exposing the onboard system to the open internet.

Responsibility for the task

The person responsible must establish and document communication via a 5G modem and Husarnet VPN and investigate weak coverage, interruptions, recovery and access boundaries.

The task is complete when

- The boat can be reached from an approved computer on shore.
- Required ROS 2 data can be exchanged in a controlled manner.
- Only necessary services are available.
- Connection loss and recovery have been tested.
- The setup can be reproduced from the documentation.

16. Operator view in Foxglove

Need and purpose

The operator must quickly be able to determine where the boat is, what it is doing and whether a test can continue safely.

Responsibility for the task

The person responsible must clarify what information safe operation requires and develop a prioritised, coherent Foxglove view.

The task is complete when

- The main view shows position, heading, route, mode, emergency stop, destination, communication and critical faults.
- Camera images and detections can be displayed.
- Critical information is clear.
- The setup works with live data and recordings.
- The configuration is stored in the repository.

17. Recording and replaying test data

Need and purpose

Faults on the water are costly and difficult to reproduce. The team needs a "black box."

Responsibility for the task

The person responsible must determine which data are stored, how recording is controlled and how a test is replayed for analysis.

The task is complete when

- Important sensor, status, command and response data are stored.
- Recordings include time and test context.
- An event can be replayed in Foxglove.

- Storage requirements and retention are documented.

8 Platform and development work

18. Reproducible Jetson AGX setup

Need and purpose

The onboard computer must start and run the software identically and predictably every time.

Responsibility for the task

The person responsible must make the Jetson AGX a reproducible runtime platform with dependencies, SSH access, project code, configuration and service startup.

The task is complete when

- A new or reinstalled device can be configured from instructions.
- The repository and dependencies can be installed without verbal assistance.
- The software starts automatically or with one documented command.
- The version and configuration can be identified.
- Startup failures are visible.

19. Shared development environment

Need and purpose

Everyone must be able to build, run and test the project without incompatible personal setups.

Responsibility for the task

The person responsible must create and maintain shared guidance for WSL2, Ubuntu 24.04 LTS, Visual Studio Code, ROS 2 and the dependencies. Cloning the repository is included here.

The task is complete when

- A new member can go from an empty computer to a running project by following the guide.
- The setup has been tested by someone other than the author.
- Common errors are documented.
- The environment resembles the Jetson setup where practical.

20. Automated code quality control

Need and purpose

The team needs prompt notification when a change breaks the build, tests or coding standards.

Responsibility for the task

The person responsible must ensure that GitHub automatically checks changes before they are merged into the main branch. The first goal is continuous integration; automatic deployment to the boat should wait until a safe release process exists.

The task is complete when

- The project is built automatically for pull requests.
- Agreed tests and checks are run.
- Failures provide understandable feedback.
- The checks are fast enough for daily use.
- The process for adding new tests is documented.

21. Protected and reviewable GitHub workflow

Need and purpose

The main branch must represent a version that the team has under control while remaining easy to contribute to.

Responsibility for the task

The person responsible must establish a simple workflow for branches, pull requests, review and merging.

The task is complete when

- Direct pushes and force pushes to the main branch are blocked.
- Changes require a pull request and the agreed checks.
- Relevant changes require approval from another person.
- The workflow is documented.
- Urgent changes can be made without losing traceability.

22. Monitoring system health

Need and purpose

Autonomous operation must not continue if required sensors, connections or programs fail.

Responsibility for the task

The person responsible must give the boat an overall understanding of technical health and, together with other responsible parties, determine which components are critical in each mode.

The task is complete when

- The system detects missing, outdated or invalid data.
- Critical faults prevent or interrupt autonomous operation.
- Faults are available to safety, lights, logging and Foxglove.
- The operator can see which component triggered the fault.

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